

Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/CLSRComputer Law
&
Security Review

On defense of “ethification” of law: How ethics may improve compliance with the EU digital laws

Matúš Mesarčík^{a,b,*}, Juraj Podroužek^b, Adrián Gavorník^b

^a Comenius University in Bratislava, Faculty of Law, Slovakia, 810 00 Bratislava

^b Kempelen Institute of Intelligent Technologies, Slovakia, 811 09 Bratislava

ARTICLE INFO

Keywords:

Ethics

Law

Artificial intelligence

Data protection

Impact assessment

ABSTRACT

In recent years, academics and professionals witness the rise of the “ethification” of law, specifically in the area of ICT law. Ethification shall be understood as a proliferation of moral principles and moral values in the legal discourse within the areas of research, innovation governance, or directly enforceable rules in the industry. Although the ethical considerations may seem distant from mere regulatory compliance, the opposite is true. The article focuses on the positive side of the “ethification” of digital laws through the lens of legal requirements for impact assessments pursuant to General Data Protection Regulation and conformity assessments in the proposal for the Artificial Intelligence Act. Authors argue that ethical considerations are often absent in the context of using new technologies including artificial intelligence, yet they may provide additional value for organizations and society as a whole. Additionally, carrying out ethics-based assessments is already in line with existing regulatory requirements in the fields of data protection law and proposed EU AI regulation. These arguments are reflected in the context of facial recognition technology, where both data protection impact assessment under the EU General Data Protection Regulation and conformity assessment under the proposal of the EU Artificial Intelligence Act will be mandatory. Facial recognition technology is analyzed through the ethics-based assessment involving stakeholder analysis, data flows map, and identification of risks and respective countermeasures to show additional insights that ethics provides beyond regulatory requirements.

© 2023 Matúš Mesarčík, Juraj Podroužek, Adrián Gavorník. Published by Elsevier Ltd. All rights reserved.

1. Introduction

In 2016, the emergence of the augmented reality game Pokémon GO sparked the interest of many players around the world.¹ The basic premise of the game lies in collecting myth-

ical characters known as Pokémon, which are located in real-world premises and may be “caught” via mobile application. To conduct a successful “catch,” a player needs special equipment - a pokéball that is available at pokéstops. Pokémon may be further trained at virtual gyms. However, as research shows, pokéstops and some Pokémon were not available in

* Corresponding author at: Comenius University in Bratislava, Faculty of Law, Slovakia, 810 00 Bratislava, &, Ethics and Law Specialist, Kempelen Institute of Intelligent Technologies, Slovakia, 811 09 Bratislava.

E-mail addresses: matus.mesarcik@kinit.sk (M. Mesarčík), juraj.podrouzek@kinit.sk (J. Podroužek), adrian.gavornik@intern.kinit.sk (A. Gavorník).

¹ Samuel Gibbs, “Pokémon Go accounts are being sold for thousands of pounds on eBay”, The Guardian (19 July 2016), <<https://www.theguardian.com/technology/2016/jul/19/pokemon-go-accounts-sold-thousands-ebay>> accessed 25.10.2022.

neighborhoods mainly inhabited by minorities.² The finding led to a discussion about whether the game is racially biased.³ Despite the fact that compliance with legal obligations might have been reached,⁴ the case reflects a deeper issue that might not be achieved through plain compliance with legislation as a potential bias of the algorithm behind the game was discovered after the launch. We are of the opinion that prior assessments might be of the essence to mitigate such risks as in the case of Pokémon GO. This is where ethics plays a crucial role.

Fields such as Science and Technology Studies (STS) have a long history of examining the nuances of the relationship between technologies and society. Recently, more legal scholars are starting to embrace some of the ideas and concepts developed within this field of research. For example, the notion of “boundary work” has been used to identify and analyze the discursive work that is put into e.g. policy documents to demarcate and claim the authority of the field of ethics as compared to the law.⁵ This is highly relevant for penetrating ethics by law and vice versa. The relationship between ethics (morality) and law is subject to rich academic debates,⁶ probably one of the most famous being a discussion between Lon L. Fuller and H. L. A. Hart published in Harvard Law Review in 1958.⁷ The debate in question represents classical dispute on the nature of law, where Hart as a legal positivist argued for separation of law and morality. On the other hand, Fuller

promotes the view that law has a moral dimension that is inseparable from norms.

However, what is currently at the center of the attention of academics and policymakers is the following setting - enforcement of ethical rules through regulation and perception of law and ethics as intertwined spheres.⁸ Some authors labeled this phenomenon as the “ethification of ICT governance,” specifically in information and communication technology regulation.⁹ The ethification of ICT governance is demonstrated as the proliferation of moral principles and moral values in the legal discourse. Specifically, ethification in this context means the invocation of soft forms of regulation through compliance with ethical standards and by imposing ethical rules; innovation governance ethics as a means of governance of technological innovation e.g. via expert groups and their guidelines;¹⁰ or industry ethics. Unlike research ethics¹¹ or innovation governance ethics, industry ethics aims to foster ethical rules directly in the industry through various tools including certification, standardization, or risk assessments.¹² This is especially true in the case of digital ethics concerning emerging technologies including artificial intelligence. Floridi even argues that digital ethics and digital regulation shape digital governance as such. Specifically, digital regulation influences digital governance through legal compliance, and simultaneously digital ethics shapes digital governance and regulation via moral evaluation.¹³

These phenomena are visible, especially in terms of data protection law and regulation of artificial intelligence (AI). The role of ethics in the EU data protection law has been emphasized by former European Data Protection Supervisor (EDPS) Buttarelli. In his speech, he was stressing its importance especially in interpreting data protection laws.¹⁴ Floridi primarily highlights the role of (soft) ethics in terms of interpreting recitals accompanying binding articles of the General Data Protection Regulation.¹⁵ Rochel even conceptualizes ethics as embedded in the EU data protection law.¹⁶ Hijmans and Raab

² Kooragayala Shiva, Srinu Tanaya, “Pokemon GO is changing how cities use public space, but could it be more inclusive?”, Urban Institute (2 August 2016), <<https://www.urban.org/urban-wire/pokemon-go-changing-how-cities-use-public-space-could-it-be-more-inclusive>> accessed 25.10.2022.

³ Ibid; or Nicole Orttung, “Is Pokemon Go racially biased?”, The Christian Science Monitor (10 August 2016), <<https://www.csmonitor.com/Technology/2016/0810/Is-Pokemon-Go-racially-biased>> accessed 25.10.2022.

⁴ Although privacy concerns were expressed soon after the launch of the game. See e.g. C. Braghin and M. Del Vecchio, “Is Pokemon GO Watching You? A Survey on the Privacy-Awareness of Location-Based Apps’ Users,” 2017 IEEE 41st Annual Computer Software and Applications Conference (COMPSAC), 2017, pp. 164–169.

⁵ Calo, Ryan, The Scale and the Reactor (April 9, 2022). Available at SSRN: <https://ssrn.com/abstract=4079851> or <http://dx.doi.org/10.2139/ssrn.4079851>.

⁶ See notably Hans Kelsen, “Reine Rechtslehre: Einleitung in die rechtswissenschaftliche Problematik,” (1934) Leipzig: F. Deuticke; Gustav Radbruch, “Gesetzliches Unrecht und übergesetzliches Recht” (1946), in *Gustav Radbruch, Gesamtausgabe*, vol. 3, ed. Arthur Kaufmann (Heidelberg: C. E. Müller, 1990), pp. 83–89; Robert Alexy, “The Argument from Injustice: A Reply to Legal Positivism,” (2020) Oxford: Oxford University Press; Ronald Dworkin, “Justice for Hedgehogs” (2011) Cambridge, Mass.: Belknap Press of Harvard University Press; Joseph Raz, “Ethics in Public Domain: Essays in the Morality of Law and Politics” (1994) Oxford: Oxford University Press, 1994; John Rawls, “A Theory of Justice”, rev. ed. (1999) Cambridge, Mass.: The Belknap Press of Harvard University Press; Jürgen Habermas, “Between Facts and Norms: Contributions to a Discourse Theory of Law and Democracy” (1996) Cambridge, Mass.: MIT Press; John Finnis, “Natural Law and the Ethics of Discourse,” *Ratio Juris* 12 (1999), pp. 354–73.

⁷ H. L. A. Hart, “Positivism and the Separation of Law and Morals” (1958) *Harvard Law Review*. 71 (4), pp. 593–629 and Lon L. Fuller, “Positivism and Fidelity to Law — A Reply to Professor Hart” (1958) *Harvard Law Review*. 71 (4), pp. 630–672.

⁸ J. Rochel, “Ethics in the GDPR: A Blueprint for Applied Legal Theory” (2021) *International Data Privacy Law*, 11(2), pp. 209–223.

⁹ Niels van Dijk, Simone Casiraghi, Serge Gutwirth, “The ‘Ethification’ of ICT Governance. Artificial Intelligence and Data Protection in the European Union,” *Computer Law & Security Review*, Volume 43, 2021.

¹⁰ Ibid.

¹¹ See e.g. discussion in Charles D. Raab, “Information privacy, impact assessment, and the place of ethics,” *Computer Law & Security Review* Volume 37, (2020), 105404, <<https://doi.org/10.1016/j.clsr.2020.105404>>.

¹² Niels van Dijk, Simone Casiraghi, Serge Gutwirth, “The ‘Ethification’ of ICT Governance. Artificial Intelligence and Data Protection in the European Union,” *Computer Law & Security Review*, Volume 43, 2021.

¹³ L. Floridi, “Soft Ethics and the Governance of the Digital” (2018) *Philos. Technol.* 31, p. 3.

¹⁴ G. Buttarelli, “Choose Humanity: Putting Dignity Back into Digital - Opening Speech of Public Session, European Data Protection Supervisor (24 October 2018), <https://edps.europa.eu/data-protection/our-work/publications/speeches-articles/choose-humanity-putting-dignity-back_en> accessed 25.10.2022.

¹⁵ L. Floridi, “Soft Ethics: Its Application to the General Data Protection Regulation and Its Dual Advantage,” (2018) *Philos. Technol.* 31, 163–167.

¹⁶ Supra n8.

argue for ethical dimensions in different balancing exercises and requirements foreseen by GDPR. Ethics also plays a crucial role in implementation of the principle of accountability through the lens of wider corporate social responsibility.¹⁷

The same applies to the regulation of AI in the EU. Early strategic documents developed by the European Commission called for an appropriate ethical and legal framework for AI.¹⁸ The call materialized in formation of a High-level expert group on artificial intelligence (AI HLEG) that issued Ethics Guidelines for Trustworthy AI (EGTAI), where the trustworthy AI is deemed not only as lawful and robust but also as ethical.¹⁹ These guidelines were later transformed into a practical tool that translates the guidelines into an accessible checklist - Assessment List for Trustworthy Artificial Intelligence (ALTAI).²⁰

Furthermore, the European Commission introduced a proposal for the first AI regulation (Artificial Intelligence Act, AIA) in the EU in the spring of 2021. In general, the ethical side of AI systems is emphasized in the text of the act.²¹ Several areas covered by ALTAI and the guidelines are presented in the AI act as well. Furthermore, ALTAI is acknowledged as the state-of-the-art minimum requirement for conformity assessment.²² The role of ethics is therefore essential in case of future compliance with the regulation. However, Clarke criticized that ethical considerations form a weaker component of the proposed regulation as the proposal does not follow the previous work of the AI HLEG.²³

¹⁷ H. Hijmans & C. Raab, "Ethical Dimensions of the GDPR, AI Regulation, and Beyond", (2022) *Direito Público*, 18(100). <<https://doi.org/10.11117/rdp.v18i100.6197>>.

¹⁸ European Commission, "Artificial Intelligence for Europe," Communication from the Commission to the European Parliament, the European Council, the Council, the European Economic and Social Committee, and the Committee of the Regions, (2018) <<https://ec.europa.eu/digital-single-market/en/news/communication-artificial-intelligence-europe>> accessed 25.10.2022.

¹⁹ High-level expert group on artificial intelligence, Ethics Guidelines for Trustworthy AI, (8 April 2019), <<https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>> accessed 25.10.2022.

²⁰ European Commission, "Assessment List for Trustworthy Artificial Intelligence (ALTAI) for self-assessment," (17 July 2020) <<https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment>> accessed 25.10.2022.

²¹ AIA, Recital 5: "By laying down those rules, this Regulation supports the objective of the Union of being a global leader in the development of secure, trustworthy, and ethical artificial intelligence, as stated by the European Council, and it ensures the protection of ethical principles, as specifically requested by the European Parliament." Recital 16: "Research for legitimate purposes in relation to such AI systems should not be stifled by the prohibition, if such research does not amount to use of the AI system in human-machine relations that exposes natural persons to harm and such research is carried out in accordance with recognized ethical standards for scientific research."

²² European Commission, "Impact Assessment of the Regulation on Artificial intelligence," (21 April 2021), <<https://digital-strategy.ec.europa.eu/en/library/impact-assessment-regulation-artificial-intelligence>> accessed 25.10.2022.

²³ R. Clarke, "The EC's Proposal for Regulation of AI: Evaluation against a Consolidated Set of 50 Principles" (2021) Review Draft

This article is a direct response to the public invitation presented in the comment "The 'Ethification' of ICT Governance. Artificial Intelligence and Data Protection in the European Union" to defend ethical practices.²⁴ We argue in favor of the positive side of the "ethification" of law and emphasize that fitting ethics considerations into existing regulatory requirements concerning data protection laws and regulation of AI is in line with what is required by law. Furthermore, such an approach may improve trustworthiness in technologies and sensitization of AI providers as formal compliance with the law does not necessarily accommodate essential societal risks.²⁵ This argument is reflected in the specific use case of facial recognition technology using our methodology for ethics-based assessments.

Our main research question is framed as follows: "How may ethics improve compliance with the selected EU digital laws by complementing existing assessments." The focus is on digital ethics and specifically on "AI ethics".²⁶ Our research is narrowed down to assessments explicitly provided by regulation in the fields of data protection and AI²⁷ in the EU. The article starts by considering different types of impact or conformity assessments with closer elaboration on the fields of EU data protection law (General Data Protection Regulation) and AI (forthcoming Artificial Intelligence Act). We will analyze the legal requirements for conducting such assessments. This part of the article closes with a landscape of AI ethics-based assessments and a description of our approach using ALTAI with certain amendments. The next part of the arti-

of 22 August 2021, <<http://www.rogerclarke.com/DV/AIP-EC21.html>> accessed 25.10.2022.

²⁴ Niels van Dijk, Simone Casiraghi, Serge Gutwirth, The 'Ethification' of ICT Governance. Artificial Intelligence and Data Protection in the European Union, *Computer Law & Security Review*, Volume 43, 2021, p. 13.

²⁵ This discussion is already covered by existing research. Ethical dimensions of GDPR and AIA are analyzed in H. Hijmans & C. Raab, "Ethical Dimensions of the GDPR, AI Regulation, and Beyond", (2022) *Direito Público*, 18(100). <<https://doi.org/10.11117/rdp.v18i100.6197>>. Different types of assessments are covered e.g. by David Wright, Michael Friedewald, "Integrating privacy and ethical impact assessments" *Science and Public Policy*, Volume 40, Issue 6, December 2013, pp. 755-766, <<https://doi.org/10.1093/scipol/sct083>>; Alessandro, Mantalero, "Beyond DataHuman Rights, Ethical and Social ImpactAssessment in AI," (2022) T.M.C.ASSER PRESS, The Hague, The Netherlands; Charles D. Raab, "Information privacy, impact assessment, and the place of ethics," *Computer Law & Security Review* Volume 37, July 2020, 105404, <<https://doi.org/10.1016/j.clsr.2020.105404>>; David, Wright - Charles D. Raab, "Constructing a surveillance impact assessment" *Computer Law & Security Review* Volume 28, Issue 6, December 2012, pp. 613-626.

²⁶ Our perception of digital ethics fits into the concept of industry ethics mentioned above with a focus on the use of digital technologies through compliance requirements of impact assessments.

²⁷ The analysis and approach presented in this paper is aimed to consider explicit legal requirements also in the context of proposed AI act and obligation to conduct conformity assessment therefore beyond integration of data protection (privacy) impact assessment and ethics-based assessment. See e.g. David Wright - Emilio Mordini, "Privacy and Ethical Impact Assessment. In D. Wright - P. De Hert editors. *Privacy impact assessment*. Dordrecht: Springer; 2012, pp. 397-419.

cle introduces the use case of facial recognition technologies that are subject to the requirement to conduct data protection impact assessment pursuant to GDPR and conformity assessment according to AIA. After initial legal considerations, we will apply the ethics-based assessment to the case to reflect on the benefits of ethics going beyond regulation.

2. Impact and conformity assessments in law and ethics

Ex-ante assessments are not novel when it comes to EU legislation but may differ in methodology and purposes.²⁸ The area of product regulation is inherently connected to conducting conformity assessments. In practice, this means that when a manufacturer intends to place a product on the EU market, it has to undergo a conformity assessment to ensure compliance with requirements prescribed by law. In some cases, the notified authorities are involved in the process to ensure proper compliance with regulatory requirements.²⁹ Similar procedures also apply to medical devices based on the regulation of medical devices.³⁰

Impact assessments are also part of the EU law. In this paper, we are not discussing regulatory impact assessment as an obligation for the legislature to prove the significance of proposed laws (regulatory impact assessments or "RIA"). Instead, we are focusing on ex-ante impact assessments like fundamental rights assessments or environmental impact assessments (EIA).³¹ The fundamental rights impact assessments are already part of the binding legislation in the case of EU data protection law.³² The purpose of such assessment is to foster accountability of various entities and alter internal processes to mitigate risks to fundamental rights and freedoms. They also function as an "early warning system" when deploying an emerging technology into practice.³³

²⁸ For elaborate discussion on different kinds of impact assessments see e.g. Charles D. Raab, "Information privacy, impact assessment, and the place of ethics," *Computer Law & Security Review* Volume 37, July 2020, 105404.

²⁹ For more details see European Commission, "Conformity Assessment," <https://single-market-economy.ec.europa.eu/single-market/goods/building-blocks/conformity-assessment_en> accessed 25.10.2022.

³⁰ Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC

³¹ See in particular "EIA Directive." Directive 2011/92/EU of the European Parliament and of the Council of 13 December 2011 on the assessment of the effects of certain public and private projects on the environment.

³² See in particular GDPR, Article 35 and Alessandro, Mantalero, "Beyond DataHuman Rights, Ethical and Social ImpactAssessment in AI," (2022) T.M.C.ASSER PRESS, The Hague, The Netherlands.

³³ Kloza, D., van Dijk, N., Gellert, R., B"orocz, I., Tanas, A., Mantovani, E., Quinn, P., "Data protection impact assessments in the European Union: Complementing the new legal framework towards more robust protection of individuals" (2017) (1 ed.) d.pia.lab Policy Brief.

2.1. Data protection impact assessment in the general data protection regulation

The right to data protection is enshrined as a fundamental right in Article 8 of the Charter of Fundamental Rights of the EU. Further specifics and rules for the processing of personal data are covered by secondary legislation, where the most prominent space is reserved for General Data Protection Regulation (GDPR). In general, GDPR applies to the automatic and manual processing of personal data. In practice any information processing³⁴ about identified or identifiable subjects triggers the scope of GDPR.³⁵ The burden of compliance with EU data protection lies primarily on the shoulders of controllers³⁶ determining purposes of processing and partial processors³⁷ processing personal data on behalf of controllers.³⁸

Fundamentals of personal data processing are enshrined in the Article 5 of GDPR provisioning basic principles of processing. The proactive and preemptive obligations of controllers are explicitly stated in the principle of accountability.³⁹ This principle reflects the need for high engagement with data protection law and requires a proactive approach instead of formal compliance and documentary obligations.⁴⁰ The pertinent principle materializes in several general⁴¹ and specific obligations in the GDPR.^{42,43}

³⁴ GDPR, Article 4 (2): "processing means any operation or set of operations which is performed on personal data or sets of personal data, whether or not by automated means, such as collection, recording, organization, structuring, storage, adaptation or alteration, retrieval, consultation, use, disclosure by transmission, dissemination or otherwise making available, alignment or combination, restriction, erasure or destruction".

³⁵ See GDPR, Article 4 (1): "personal data" means any information relating to an identified or identifiable natural person ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person" as interpreted by the decisions of the Court of Justice of the European Union in cases C-434/16 *Nowak* and C-582/14 *Patrick Breyer*.

³⁶ GDPR, Article 4 (7).

³⁷ GDPR, Article 4 (8).

³⁸ For deeper analysis see Brendan Van Alsenoy, "Allocating responsibility among controllers, processors, and "everything in between": the definition of actors and roles in Directive 95/46/EC" (2012), *Computer Law and Security Review*. 2012, Vol. 28, No. 1.

³⁹ GDPR, Article 5 (2)

⁴⁰ Article 29 Data Protection Working Party, "Opinion 3/2010 on the principle of accountability," adopted on 13 July 2010. 00062/10/EN WP 173. Joseph Alhadeff, Brendan Van Alsenoy and J. Dumortier, "The accountability principle in data protection regulation: origin, development and future directions" In D. Guagnin, L. Hempel, C. Ilten a.o. (eds.), *Managing Privacy through Accountability*, 2012, Palgrave Macmillan, p. 49 - 82.

⁴¹ See GDPR, Article 24.

⁴² Including requirements for designation of Data Protection Officer under Article 37, data protection by design and by default under Article 25 or conducting Data Protection Impact Assessments (DPIA) under Article 35.

⁴³ Joseph Alhadeff, Brendan Van Alsenoy and J. Dumortier, "The accountability principle in data protection regulation: origin, development and future directions" In D. Guagnin, L. Hempel, C. Ilten

2.1.1. Processing operations subject to DPIA

DPIA as provisioned in Article 35 of GDPR has been often described as a crucial part of the accountability principle.⁴⁴ In the words of Article 29 Data Protection Working Party (WP29)⁴⁵ it “is a process designed to describe the processing, assess the necessity and proportionality of a processing and to help manage the risks to the rights and freedoms of natural persons resulting from the processing of personal data (by assessing them and determining the measures to address them).”⁴⁶ This in practice means that the institution of DPIA “enhances compliance” with GDPR in high-risk cases of personal data processing.⁴⁷

However, from the point of applicability of the DPIA, not every processing operation of personal data requires conducting DPIA (although a controller may assess their processing operations voluntarily). The obligation to conduct DPIA may stem from three sources. Firstly, the DPIA is mandatory in scenarios foreseen by Article 35 (3) GDPR.⁴⁸ Secondly, the obligation to conduct DPIA stems from the “general clause” of Article 35 (1) “where a type of processing, in particular, using new technologies, and taking into account the nature, scope, context, and purposes of the processing, is likely to result in a high risk to the rights and freedoms of natural persons.”⁴⁹ Analysis of the notion of “high-risk” is of the essence here as not every processing operation results in a high risk to the rights and freedoms of data subjects. The risk analysis in the EU data protection law should be assessed in a broader context and methodologically different than in other areas of regulation.⁵⁰ It is argued that the risk analysis within DPIA encompasses the triple assessment of what exactly constitutes

the risk, the severity of the risk, and the likelihood of the risk.⁵¹

WP29 in its guidelines lists 10 criteria that should be considered within the analysis if processing operations lead to a “high-risk.”⁵² The more criteria are met, the higher probability of operations occurring at high risk. According to WP29, as a rule of thumb, if two criteria are met, DPIA may be needed.⁵³ Literature provides different methodologies for quantifying risks in terms of DPIA.⁵⁴

Thirdly, DPIA is an obligatory requirement when a processing operation is listed by data protection authorities as requiring DPIA.⁵⁵ Additionally, member states may require conducting DPIA in specific cases.⁵⁶

2.1.2. The content of the DPIA

Article 35 (7) of GDPR reflects what precisely shall be conducted within DPIA. According to this article, the DPIA shall contain at least four aspects. Firstly, the assessment shall contain “a systematic description of the envisaged processing operations and the purposes of the processing, including, where applicable, the legitimate interest pursued by the controller.”⁵⁷ Following this requirement, the controller shall set out the nature, scope, context, and purposes of processing that are subject to DPIA.⁵⁸ Processing operations shall be described together with a specification of personal data used, recipients and data retention periods, and assets used for storage.⁵⁹ Secondly, DPIA shall contain an assessment of necessity and proportionality taking into account the purposes of the processing.⁶⁰ According to WP29, this step should evaluate specific measures foreseen by GDPR and design to comply with basic principles of processing or individual rights management including determining data retention periods, legitimate interests, data minimization, safeguards surrounding the cross-

a.o. (eds.), *Managing Privacy through Accountability*, 2012, Palgrave Macmillan, p. 49 - 82.

⁴⁴ Article 29 Data Protection Working Party, “Guidelines on Data Protection Impact Assessment (DPIA) and determining whether the processing is “likely to result in a high risk” for the purposes of Regulation 2016/679” Adopted on 4 April 2017 As last Revised and Adopted on 4 October 2017, WP 248 rev.01; Katerina Demetrou, “Data Protection Impact Assessment: A tool for accountability and the unclarified concept of ‘high risk’ in the General Data Protection Regulation,” *Computer Law & Security Review*, Volume 35, Issue 6, 2019.

⁴⁵ Now the European Data Protection Board (EDPB).

⁴⁶ Article 29 Data Protection Working Party, “Guidelines on Data Protection Impact Assessment (DPIA) and determining whether the processing is “likely to result in a high risk” for the purposes of Regulation 2016/679” Adopted on 4 April 2017. As last Revised and Adopted on 4 October 2017, WP 248 rev.01, p. 3.

⁴⁷ GDPR, Recital 84.

⁴⁸ GDPR, Article 35 (3) lists (i) “a systematic and extensive evaluation of personal aspects relating to natural persons which are based on automated processing, including profiling, and on which decisions are based that produce legal effects concerning the natural person or similarly significantly affect the natural person; (ii) processing of special categories of data referred to in Article 9(1) on a large scale, or processing of personal data relating to criminal convictions and offenses referred to in Article 10; or (iii) a systematic monitoring of a publicly accessible area on a large scale”

⁴⁹ GDPR, Article 35 (1).

⁵⁰ Van Dijk, N., Gellert, R. and Rommetveit, K. “A Risk to a Right: Beyond Data Protection Risk Assessments” (2016) Volume 32, Issue 2, *Computer Law & Security Review*, pp. 286-306.

⁵¹ Katerina Demetrou, “Data Protection Impact Assessment: A tool for accountability and the unclarified concept of ‘high risk’ in the General Data Protection Regulation,” *Computer Law & Security Review*, Volume 35, Issue 6, 2019, p. 5.

⁵² These criteria include evaluation or scoring including profiling and predicting; automated-decision making with legal or similar significant effect; systematic monitoring; the processing of special categories of personal data as enshrined in the Article 9 GDPR; personal data processed on a large scale; matched or combined datasets; the processing of personal data of vulnerable persons; innovative use or application of technological and organizational solutions; cross-border transfer of personal data outside the EU and processing that is preventing data subjects from exercising a right or using a service or a contract. Article 29 Data Protection Working Party, “Guidelines on Data Protection Impact Assessment (DPIA) and determining whether the processing is “likely to result in a high risk” for the purposes of Regulation 2016/679.” Adopted on 4 April 2017. As last Revised and Adopted on 4 October 2017, WP 248 rev.01, p. 8-9.

⁵³ Ibid, p. 9.

⁵⁴ Supra n51, p. 8 ff.

⁵⁵ GDPR, Article 35 (4).

⁵⁶ For example in Slovakia, DPIA was required for state-developed quarantine monitoring mobile applications.

⁵⁷ GDPR, Article 35 (7) (a).

⁵⁸ GDPR, Recital 90.

⁵⁹ Supra n56, p. 21.

⁶⁰ GDPR, Article 35 (7) (b).

border flow of personal data, and other.⁶¹ Thirdly, the controller shall assess the risks to the rights and freedoms of data subjects.⁶² The assessment should contain a comprehensive analysis of potential impacts, threats, likelihood and severity, and risk sources taking into account risky scenarios for data subjects.⁶³ This shall be interpreted as the most crucial part of the DPIA resembling fundamental rights assessments as specific rights, freedoms, and interests of data subjects are under evaluation for potential harm and risks. The fourth step of DPIA should describe and implement mitigating measures for the risks identified in the previous part.⁶⁴ It is of the essence to note, that where appropriate, the controller “shall seek the views of data subjects or their representatives on the intended processing, without prejudice to the protection of commercial or public interests or the security of processing operations.”⁶⁵ As the provision indicates, controllers might escape the requirement either by arguing that the survey would be inappropriate or that such activity would be a heavy burden from the point of finances, public interests, or security. However, as emphasized by WP29,⁶⁶ controllers may seek the views of data subjects by variable means (studies, surveys, forms) therefore limiting the space for maneuvering in non-compliance with the requirement in question.

2.2. Conformity assessments pursuant to the AI act

The proposal for the Artificial Intelligence Act (AI act or AIA) introduced by the European Commission in the spring of 2021 aims to provide the EU with the first comprehensive regulation of AI.⁶⁷ The proposal represents a horizontal regulation requiring a risk-based approach and builds on the current system of regulation of products and services in the EU. AIA defines AI systems as “software that is developed with one or more of the techniques and approaches listed in Annex I and can, for a given set of human-defined objectives, generate outputs such as content, predictions, recommendations, or decisions influencing the environments they interact with.”⁶⁸ Additionally, Annex I lists techniques and approaches applicable to the definition of AI systems.⁶⁹ The definition has been subject to criticism by academics and practitioners due to high reliance on the term AI in general, potential extensive scope, differ-

ences and interpretation of technical approaches and considering capabilities.⁷⁰ It is expected that the final version of the AI act will reflect the criticism and define AI systems in line with current technological state-of-the-art.⁷¹

The AI act proposes four categories of AI systems based on their risk for fundamental rights, health, and safety.⁷² On the top, the category of banned practices is enshrined in Article 5 of AIA. Additional obligations are placed for ‘real-time’ and ‘post’ remote biometric identification systems in publicly accessible spaces for the purpose of law enforcement. AIA is primarily focused on setting obligations for the category of high-risk AI systems, either standalone high-risk AI systems including biometric categorization or law enforcement or AI systems as a safety component, or products covered by the EU harmonization legislation.⁷³ The European Commission has the power to amend Annex III by delegated acts to dynamically reflect technological development.⁷⁴ Further requirements are applicable to AI systems posing limited risks, especially from the point of transparency obligations⁷⁵ and providers of AI systems with minimal or no risks are encouraged to draft and implement a specific code of conduct.⁷⁶

2.2.1. The scope of conformity assessment in the AI act

As pointed out above, specific requirements are primarily aimed at the providers (or in specific cases product manufacturers, distributors, importers, or third parties) of high-risk AI systems. Prior to placing a high-risk AI system on the market, the provider is obliged to undergo a conformity assessment (CA) that is a part of auditing mechanisms provided by the AIA. Various AI auditing tools and methods are not novel⁷⁷ and in the context of AI governance may serve different purposes and tasks.⁷⁸

⁷⁰ See e.g. discussion in J. Schuett, “Defining the scope of AI regulations,” (2023) *Law, Innovation and Technology*, 0(0), pp. 1–23, <<https://doi.org/10.1080/17579961.2023.2184135>> or Matúš Mesarčík, Sára, Soľárová, Juraj Podroužek, Mária Bieliková “Stance on the Proposal for a Regulation of the European Parliament and of the Council Laying down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act) and Amending Certain Union Legislative Acts,” (2021), <<https://kinit.sk/publication/stance-on-the-proposal-for-a-regulation-laying-down-harmonised-rules-on-artificial-intelligence-artificial-intelligence-act/>>, accessed 30.5.2023.

⁷¹ The European Parliament proposed in its position towards AIA alignment with definition of AI according to OECD. See Luca, Bertuzzi, “EU lawmakers set to settle on OECD definition for Artificial Intelligence,” <<https://www.euractiv.com/section/artificial-intelligence/news/eu-lawmakers-set-to-settle-on-oecd-definition-for-artificial-intelligence/>>, accessed 30.5.2023.

⁷² AIA, Recital 1.

⁷³ What constitutes a high-risk AI system is defined by the proposed Annex II (for AI systems as safety components already covered by specific EU legislation) and Annex III of AIA via specific areas and applications.

⁷⁴ AIA, Article 7 (1). It shall be also noted that specific areas of high-risk are still under consideration in the legislative process.

⁷⁵ AIA, Article 53.

⁷⁶ AIA, Article 69.

⁷⁷ J. Simko, M. et al., “Towards Continuous Automatic Audits of Social Media Adaptive Behavior and its Role in Misinformation Spreading,” (2021) 29th ACM Conference on UMAP 21.

⁷⁸ Auditing tools might be used to evaluate compliance with legal standards by regulators and supervisory authorities, mitigate

⁶¹ Supra n52, p. 21.

⁶² GDPR, Article 35 (7) (c).

⁶³ Supra n52, p. 21.

⁶⁴ GDPR, Article 35 (7) (d).

⁶⁵ GDPR, Article 35 (9).

⁶⁶ Supra n52, p. 13.

⁶⁷ Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL LAYING DOWN HARMONIZED RULES ON ARTIFICIAL INTELLIGENCE (ARTIFICIAL INTELLIGENCE ACT) AND AMENDING CERTAIN UNION LEGISLATIVE ACTS. COM/2021/206 final.

⁶⁸ AIA, Article 3 (1).

⁶⁹ Namely, (a) machine learning approaches, including supervised, unsupervised, and reinforcement learning, using a wide variety of methods including deep learning; (b) logic and knowledge-based approaches, including knowledge representation, inductive (logic), programming, knowledge bases, inference and deductive engines, (symbolic) reasoning and expert systems; (c) statistical approaches, Bayesian estimation, search and optimization methods.

CA is defined as a process “of verifying whether the requirements set out in Title III, Chapter 2 of this Regulation relating to an AI system have been fulfilled”⁷⁹ therefore fostering the role of accountability⁸⁰ in AIA by putting emphasis on self-assessment for providers (and other related subjects) of AI systems. Only in exceptional cases cooperation with designated authority will be required in conducting CA. The majority of providers of high-risk AI systems shall follow the procedure for CA based on internal control as external control is obligatory only in a limited number of cases.⁸¹ The rationale behind the obligation to conduct CA internally on the side of the provider is that providers of AI systems have more experience and understanding of the specifics.⁸² External control (third-party CA) is required only in the case of AI systems that will be operated in the area of biometric identification and categorization of natural persons such as surveillance systems in publicly accessible spaces. These systems capture biometric data without significant delay (in the case of real-time)⁸³ from “live” or “near live” material such as video footage, generated by camera or other device or various forms of ‘post’ remote biometric identification⁸⁴ of people including analyzing footage from CCTV with biometric systems. The range of external control is thus limited.⁸⁵

The content of CA shall be derived from requirements for compliance with high-risk AI systems provisioned in Chapter 2 of Title III AIA. These requirements are designed to “effectively mitigate the risks for health, safety, and fundamental rights, as applicable in the light of the intended purpose of the system.”⁸⁶ Requirements include: establishment, implementation, documentation, and maintenance of risk management systems;⁸⁷ data and data governance processes;⁸⁸ drawing up of technical documentation;⁸⁹ record-keeping;⁹⁰ design and development of transparency and information provision

to users;⁹¹ design and development of human oversight;⁹² and ensuring accuracy, robustness, and cybersecurity.⁹³

AIA also contains a presumption of compliance with these requirements. The presumption applies in cases of high-risk AI systems following relevant harmonized standards.⁹⁴ The significant role of standards-developing organizations in terms of AIA has been criticized by Veale and Borgesius for concerns related to the protection of fundamental rights as standards adopted by standardization bodies might be free from judicial scrutiny and procedural aspects of the adoption of standards including the fact that the European Parliament lacks the power to veto harmonized standards endorsed by the European Commission.⁹⁵

As noted above, CA shall be performed prior to placing a high-risk AI system on the EU market or prior to putting a high-risk system into service. Additionally, CA shall be conducted also in cases where a high-risk AI system is substantially modified. This is the case of change affecting compliance with AIA or modification of the intended purpose. Continual learning of AI systems shall not be considered a substantial modification.⁹⁶

EU member states are allowed to derogate from CA procedures in the territory of the respective member states in exceptional cases.⁹⁷

2.2.2. The content of the CA in the AI act

The content of the CA depends on the fact that it shall be conducted on the responsibility of the provider of a high-risk AI system or by a third party. In this article we primarily focus on self-assessment methods, therefore the content of internal CA shall be discussed. In this case, the content of the CA is governed by Annex VI of the AIA.⁹⁸ Three specific aspects of CA procedure based on internal control are explicitly provided by Annex VI, namely: verification of quality management system; examination of information contained in the technical documentation to verify compliance with Title III, chapter 2; verification of the consistency of the design and development process of the AI system and its post-market monitoring as referred to in Article 61 with the technical documentation.

It shall be emphasized that CA is not a “one-off exercise” similar to DPIA according to GDPR.⁹⁹ The providers shall continuously evaluate, audit and establish post-market monitor-

different kinds of risks by providers and developers of AI systems, or make informed decisions by publicstakeholders. See S. Brown, J. Davidovic & A. Hasan, “The algorithm audit: Scoring the algorithms that score us,” (2021) *Big Data & Society*, 8(1).

⁷⁹ AIA, Article 3 (20).

⁸⁰ On the role of accountability concerning data protection and AI systems see e.g. H. Hijmans & C. Raab, “Ethical Dimensions of the GDPR, AI Regulation, and Beyond”, (2022) *Direito Público*, 18(100). <<https://doi.org/10.11117/rdp.v18i100.6197>> referencing M. Abrams, J. Abrams, P. Cullen and L. Goldstein, “Artificial Intelligence, Ethics and Enhanced Data Stewardship”, (2019) *IEEE Security & Privacy*, 17(2).

⁸¹ Michael Veale, and Frederik Zuiderveen Borgesius, “Demystifying the Draft EU Artificial Intelligence Act — Analyzing the good, the bad, and the unclear elements of the proposed approach,” (2021) *Computer Law Review International*, vol. 22, no. 4, 2021.

⁸² AIA, Recital 64.

⁸³ AIA, Article 3 (37).

⁸⁴ AIA, Article 3 (38).

⁸⁵ M. Haataja, & J. J. Bryson, “What costs should we expect from the EU’s AI Act?” (2021), <<https://doi.org/10.31235/osf.io/8nzb4>>.

⁸⁶ AIA, Recital 43.

⁸⁷ AIA, Article 9.

⁸⁸ AIA, Article 10.

⁸⁹ AIA, Article 11.

⁹⁰ AIA, Article 12.

⁹¹ AIA, Article 13.

⁹² AIA, Article 14.

⁹³ AIA, Article 15.

⁹⁴ AIA, Articles 40 and 42. This might be the case of compliance with cybersecurity obligations where a statement of conformity has been issued to the provider of high-risk AI systems according to a cybersecurity scheme under specific EU regulation e.g. Regulation (EU) 2019/881 of the European Parliament and of the Council of 17 April 2019 on ENISA (the European Union Agency for Cybersecurity) and on information and communications technology cybersecurity certification and repealing Regulation (EU) No 526/2013 (Cybersecurity Act) PE/86/2018/REV/1. OJ L 151, 7.6.2019, p. 15–69.

⁹⁵ *Supra* n81, p. 105.

⁹⁶ AIA, Recital 66.

⁹⁷ AIA, Article 47 (1).

⁹⁸ AIA, Article 43 (1) (a), (2).

⁹⁹ Katerina Demetzou, “Introduction to the Conformity Assessment under the draft EU AI Act,” (12 August 2022), Available at <<https://fpf.org/blog/introduction-to-the-conformity>>.

ing plans for high-risk AI systems. Post-market monitoring systems form a fundamental pillar of AI auditing mechanisms as foreseen by AIA.¹⁰⁰

2.3. Ethics-based assessments

Ethics-based assessments resemble relatively emergent efforts in the field of AI ethics. They exist as part of a broader landscape which includes various forms of ethics frameworks ranging from guidelines, codes and standards, to practical assessment lists and expert technical tools.¹⁰¹ They also build on existing practices in assessing ethical and societal issues concerning the impacts of technology on human lives exemplified by e.g. technology assessments, social and human rights impact assessments or algorithmic assessments.¹⁰²

Because ethical frameworks and assessments can be conceived as a particular attempt to operationalize ethical principles and values such as privacy, transparency or fairness into practice, the actual effect and usefulness of such operationalization efforts are often questioned.¹⁰³ The takeaway here is not that we should abandon such efforts, but instead take the operationalization of AI ethics through frameworks more seriously and not reduce it to its instruments, tools, or checklists. The focus of conducting AI ethics-based assessments should not be on following narrow box-ticking strategy and formal compliance with some rigid set of requirements but to create a space where a sensitization of AI providers on ethical issues and ability to form sound moral judgements can be better facilitated and negotiated

There are many risks associated with the operationalization of AI ethics in the context of ethics-based assessments. One of these risks relates to the possibility of misuse or co-optation of these practices by actors who might not be genuinely interested in mitigating the ethical risks of technology. Such practices are best described by philosopher Floridi. He conceptualizes and warns of “ethics shopping”¹⁰⁴ a practice through which actors might simply pick from the existing market of principles, guidelines, or assessments to retrofit and justify their current behavior, thus avoiding any real implementation or changes. This also relates to what he calls “ethics bluwashing,”¹⁰⁵ a practice of making unsubstantiated and superficial measures by actors to make their practices appear more ethical.

assessment-under-the-draft-EU-ai-act-and-how-it-compares-to-dpias/> accessed 25.10.2022.

¹⁰⁰ J. Mokander, M. Axente, F. Casolari, et al., “Conformity Assessments and Post market Monitoring: A Guide to the Role of Auditing in the Proposed European AI Regulation,” (2022) *Minds & Machines* 32, 241–268.

¹⁰¹ Jacqui Ayling and Chapman Adriane, “Putting AI Ethics to Work: Are the Tools Fit for Purpose?” (2022) *AI and Ethics* 2 (3): 405–29. <<https://doi.org/10.1007/s43681-021-00084-x>>.

¹⁰² Supra n8.

¹⁰³ L. Munn, “The uselessness of AI ethics,” (2022) *AI Ethics*, <<https://doi.org/10.1007/s43681-022-00209-w>>.

¹⁰⁴ Floridi, L. “Translating Principles into Practices of Digital Ethics: Five Risks of Being Unethical,” (2019) *Philos. Technol.* 32, 185–193 <<https://doi.org/10.1007/s13347-019-00354-x>>.

¹⁰⁵ Ibid

While keeping these possible pitfalls in mind, by implementing the ethics-based assessment it is still possible to better operationalize values into specific practices. The ethics-based assessments and audits should be conceived as a part of a broader process that aims to “spark ethical deliberation, inform design choices, and help visualize the normative values embedded in those systems”.¹⁰⁶ At the same time, the focus of ethics-based assessment should not be solely on AI systems as isolated technical artifacts but to understand them as a part of a broader socio-technical context that can have a significant impact on people who develop, deploy, and use them and also people who are directly or indirectly affected by these systems. Such socio-technical systems consist of a combination of technical and social components as both humans and machines are necessary to make any technology work as intended.¹⁰⁷ Grounded in this understanding, there is a growing body of approaches that conceptualize the assessment process,¹⁰⁸ algorithmic harms,¹⁰⁹ or risk management¹¹⁰ as socio-technical.

Moreover, taking into account the need for sensitization of stakeholders to be able to understand the impacts of AI systems on society creates the space for experts in AI ethics to guide and facilitate AI providers to develop and deploy these systems in a fashion that makes them trustworthy and socially acceptable. From this perspective, ethics-based assessment can be understood as a facilitated process to increase the ethical sensitivity of the involved parties. This relates to the practice of being able to recognize, interpret, and act on ethically consequential decisions made within technology teams.¹¹¹

Despite the abundance¹¹² of ethical frameworks, guidelines and tools that have the ambition to operationalize ethical values and principles in AI, it is unclear how many are actually successful in achieving this. Some of them¹¹³ seem to

¹⁰⁶ J. Mokander, L. Floridi, “Operationalizing AI governance through ethics-based auditing: an industry case study,” (2022) *AI Ethics*, <<https://doi.org/10.1007/s43681-022-00171-7>>.

¹⁰⁷ Andrew, D. Selbst et al. “Fairness and Abstraction in Sociotechnical Systems,” In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, (2019), pp. 59–68. Atlanta GA USA: ACM. <<https://doi.org/10.1145/3287560.3287598>>.

¹⁰⁸ Leila Methnani, Mattias Brännström, and Andreas Theodorou, “Operationalising AI Ethics: Conducting Socio-Technical Assessment,” (2023) In *Human-Centered Artificial Intelligence: Advanced Lectures*, edited by Mohamed Chetouani, Virginia Dignum, Paul Lukowicz, and Carles Sierra, 304–21. Lecture Notes in Computer Science. Cham: Springer International Publishing. <https://doi.org/10.1007/978-3-031-24349-3_16>.

¹⁰⁹ Renee Shelby et al, “Identifying Sociotechnical Harms of Algorithmic Systems: Scoping a Taxonomy for Harm Reduction” (2023), available at arXiv <<http://arxiv.org/abs/2210.05791>>.

¹¹⁰ Elham Tabassi, “AI Risk Management Framework: AI RMF (1.0),” (2023) NIST AI 100-1. Gaithersburg, MD: National Institute of Standards and Technology. <<https://doi.org/10.6028/NIST.AI.100-1>>.

¹¹¹ K.L. Boyd, K. Shilton, “Adapting Ethical Sensitivity as a Construct to Study Technology Design Teams”, (2022) *Proc ACM Hum.-Comput. Interact.* 5, 1–29, <<https://doi.org/10.1145/3463929>>.

¹¹² See for example the OECD Catalogue of Tools & Metrics for Trustworthy AI

¹¹³ Roberto V. Zicari et al., “Z-Inspection ® : A Process to Assess Trustworthy AI,” (2021) *IEEE Transactions on Technology and Society* 2 (2): 83–97, <<https://doi.org/10.1109/TTS.2021.3066209>>

be better fit for this purpose, as they put significant attention into contextualizing the particular ethical risks, explain their theoretical underpinnings and offer some form of strategy on how to mitigate these risks. However, this also means that more effort and expertise is required to apply these frameworks into practice. For the purpose of this paper, we turn to ALTAI,¹¹⁴ a well-known framework that was developed as part of the EGTAI by the High-Level Expert Group on Artificial Intelligence. EGTAI represents an instance of a top-down approach that aims to align AI systems with the principles and requirements of trustworthy AI. The concept of trustworthy AI conforms with the idea that AI systems should be developed, deployed and used in compliance with laws, but requirements on legal compliance form only necessary but not sufficient conditions. Trustworthy AI systems should also perform in a way that will prevent unintentional harms to maintain their robustness, safety and security. And on top of that, they should be aligned with ethical principles that demand respect for human autonomy, prevention of harm, fairness and explicability.¹¹⁵

These ethical principles, derived from traditional principles of bioethics,¹¹⁶ are embodied in ALTAI to help AI providers understand the various aspects of trustworthiness and to identify possible shortcoming or problems of their systems. The requirements for trustworthy AI are not chosen arbitrarily but emanate from a broader and still ongoing debate about which principles are the most suitable to capture the essential criteria of trustworthy AI systems, with research showing a growing convergence among identified principles across documents produced by various stakeholders.¹¹⁷

Requirement for human agency and oversight deals with the effects of AI systems on human behavior, perception, or the role of humans and their oversight within the system, examining approaches such as human-in-the-loop, human-on-the-loop, or human-in-command to avoid over-reliance on AI systems. The requirement for technical robustness and safety highlights areas such as resilience to attacks and security; ac-

curacy but also reliability, fall-back plans, and reproducibility. The requirement for privacy and data governance builds upon the fundamental right to the integrity of the person, which covers respect for a person's mental and physical integrity. It discusses questions related to the impact of the AI system on individuals' privacy and adherence to various elements concerning data governance. Traceability, explainability, and open communication fall under the fourth requirement of transparency. The fifth requirement deals with the avoidance of unfair bias, accessibility, and universal bias but also stakeholder participation. Building on this, the sixth requirement considers the broader society and the environment as relevant stakeholders throughout the AI system's life cycle and discusses the impacts on environmental well-being, impact on work and skills but also on society at large and democracy as such. Last but not least, the requirement for accountability entails processes of auditability and systematic management of risks and value trade-offs.

To effectively utilize ALTAI as a tool for ethics-based assessment there are still some additional steps that need to be implemented to properly operationalize proposed aims of such assessment into practice.¹¹⁸

The first step is to analyze and identify direct and indirect stakeholders who might be affected by the AI system.¹¹⁹ In particular, identification of indirect stakeholders is often more difficult since these groups of people, or organizations are also affected by the AI system but not in a direct way or immediately after deployment of the system. It has to be stressed that affected people and organizations are not equally vulnerable regardless of taking the role of direct or indirect stakeholder. We can imagine indirect stakeholders e.g. family members or relatives of AI users, who could be at least equally affected by AI systems as some direct stakeholders.¹²⁰

To better understand various vulnerable groups in more detail the stakeholder analysis can be enriched by additional factors such as their vulnerability understood as a propensity to be affected and agency as a degree of control over a system behavior.¹²¹ Another strategy that can help AI providers to understand better the impact that AI systems pose on stakehold-

or Philip Brey, Lundgren Björn, Kevin Macnish, and Mark Ryan. n.d., "Guidelines for the Ethical Development of AI and Big Data Systems: An Ethics by Design Approach." *Shaping the Ethical Dimensions of Smart Information Systems— a European Perspective (SHERPA)*. <<https://www.project-sherpa.eu/wp-content/uploads/2019/12/development-final.pdf>>.

¹¹⁴ European Commission, "Assessment List for Trustworthy Artificial Intelligence (ALTAI) for self-assessment," (17 July 2020) <<https://digital-strategy.ec.europa.eu/en/library/assessment-list-trustworthy-artificial-intelligence-altai-self-assessment>> accessed 25.10.2022.

¹¹⁵ European Commission, and Content and Technology Directorate-General for Communications Networks. "Ethics Guidelines for Trustworthy AI," (2019) Publications Office <<https://doi.org/10.2759/177365>>.

¹¹⁶ Luciano Floridi and Josh Cowls, "A Unified Framework of Five Principles for AI in Society." (2019) *Harvard Data Science Review* 1 (1). <<https://doi.org/10.1162/99608f92.8cd550d1>>.

¹¹⁷ Jessica Fjeld, Achten Nele, Hannah Hilligoss, Adam Nagy, and Madhulika Sri Kumar, "Principled Artificial Intelligence: Mapping Consensus in Ethical and Rights-Based Approaches to Principles for AI" (2022), SSRN Scholarly Paper ID 3518482. Rochester, NY: Social Science Research Network. <<https://doi.org/10.2139/ssrn.3518482>>.

¹¹⁸ Adrian Gavornik, Juraj Podroužek, Matus Mesarcik, Sara Solárová, Stefan Oresko, and Maria Bielikova, "Utilising the Assessment List for Trustworthy AI: Three Areas of Improvement," (2022) In , 3275:8. Vienna, Austria: CEUR-WS.org. <<https://ceur-ws.org/Vol-3275/paper3.pdf>>.

¹¹⁹ Friedman, Batya, Peter H. Kahn, Alan Borning, and Alina Hultgren, "Value Sensitive Design and Information Systems" (2013) In *Early Engagement and New Technologies: Opening up the Laboratory*, edited by Neelke Doorn, Daan Schuurbijs, Ibo van de Poel, and Michael E. Gorman, 16:55–95. Dordrecht: Springer Netherlands, <https://doi.org/10.1007/978-94-007-7844-3_4>.

¹²⁰ Alexei Czeskis, Ivayla Dermendjieva, Hussein Yapit, Alan Borning, Batya Friedman, Brian Gill, and Tadayoshi Kohno, "Parenting from the Pocket: Value Tensions and Technical Directions for Secure and Private Parent-Teen Mobile Safety" (2010) In *Proceedings of the Sixth Symposium on Usable Privacy and Security*, 1–15. SOUPS '10. New York, NY, USA: Association for Computing Machinery. <<https://doi.org/10.1145/1837110.1837130>>.

¹²¹ Margarita Boyarskaya, Alexandra Olteanu, and Kate Crawford, "Overcoming Failures of Imagination in AI Infused System Development and Deployment" (2020) arXiv. <<http://arxiv.org/abs/2011.13416>>.

ers and that can lead to more tailored communication and engagement with them can be based on the estimation of the degree of foreseeable benefits and harms of the AI system on the stakeholder and the degree to which stakeholder would be aware or sensitive to foreseeable impacts. The first estimations of such degrees can be done by AI providers but ideally affected groups are to be consulted directly¹²² or included in the assessment process.

Second step is focused on depiction of data flows to better visualize and categorize the various data practices that take place during the development, deployment and use of an AI system. Current AI systems based on machine learning techniques are heavily dependent on data that feeds them. To make better sense of the flows of data from multiple sources, various tools that map data ecosystems¹²³ can be utilized. The people and organizations who participate in or benefit from data need to understand not only benefits but also potential limitations of data processing and to identify practices that can be deemed unfair by the stakeholders.

Moreover, each stage of data processing from data collection to data use and data sharing can carry unique ethical issues ranging from bias in data collection, gaps and omissions in data, to data misuse or miscommunication when sharing the data.¹²⁴ The affected stakeholders thus could be vulnerable to various degrees also depending on which stage of the data practices lifecycle they are involved in. It is therefore useful to visualize the flows of data in AI systems during their collection, use, or sharing with an emphasis on the impacts on the rights of stakeholders over data to support responsible data stewardship¹²⁵ of AI providers. Especially, identification of relevant data sources allows us also to validate affected groups identified in previous steps that are accounted for in the data collection or their capability to control the flow of their data.

Similarly, to the estimation of the degree to which direct and indirect stakeholders would be impacted, it is also possible to estimate various degrees of the urgency of ethical and societal issues that emerge by discussing ALTAI questions and recommendations. Emphasis on the risk-based approach which is already present in other AI ethics frameworks¹²⁶ can help to navigate AI providers when encountering these issues in practice. There are strategies on how to approach ethical risks in AI by using a rich apparatus of risk management methods for software and information-based systems¹²⁷ or

simply by addressing the risks by evaluating their severity and likelihood of infringement on ethical principles affected by the innovation.¹²⁸ Some approaches address a gap in the process of the initial identification of ethical risks by proposing their own questionnaires¹²⁹ to elicit these risks and to map value trade-offs.

Our empirical experience with conducting ethics-based assessments grounded in the ALTAI within industry collaborations and various research projects has shown as most useful to adopt a five-point scale to evaluate ethical risks regarding the dimensions as is the level of exposure to impacts by the most affected stakeholder groups, likelihood, and severity of impacts on stakeholders. Additionally, such evaluation shall be supplemented by various countermeasures that actors could provide by embracing various forms of direct action, indirect influence or active monitoring. We believe that the arrangement of ethical issues into risk matrices that are prepared and validated with some of the most affected stakeholder groups helps actors to better understand which issues need to be dealt with foremost while avoiding the risks of poor operationalization of principles and values of AI ethics into practice.

These specific aspects will be discussed in the next chapter in the context of an ethics-based assessment of facial recognition technology systems.

3. Assessment of facial recognition technology: the need for an interdisciplinary approach

As mentioned in the introduction, we present the specific use case of facial recognition technology (FRT) to reflect the positive aspects of the “ethification” of law. Considering the specifics of the use case, we will provide legal considerations on the application of the regulations discussed above and apply our methodology for ethics-based assessments.

FaceReco established in the European Union develops and deploys an AI system named BioID. The main purpose of BioID is the identification of individuals based on facial recognition technology in real time that might be used in different settings and spaces including controlled access to buildings, verification of voters during the elections, monitoring, and search conducted by public authorities or at the airports or stadiums for enhancement of security. BioID allows adding whitelists and blacklists of individuals, therefore, aiming at one-to-many identification. The model used for BioID was trained on publicly available pictures on the Internet and specific databases provided for commercial and non-commercial purposes online. The AI system in question uses machine learning techniques.

¹²² David, Wright - Charles D. Raab, “Constructing a surveillance impact assessment” Computer Law & Security Review Volume 28, Issue 6, December 2012, pp. 613-626.

¹²³ <https://www.theodi.org/article/data-ecosystem-mapping-tool/>

¹²⁴ Open Data Institute. “Data Ethics Canvas,” (2021) <<https://theodi.org/article/the-data-ethics-canvas-2021/>>.

¹²⁵ Open Data Institute, “Defining Responsible Data Stewardship,” (2023) <<https://www.theodi.org/article/defining-responsible-data-stewardship/>>.

¹²⁶ D. Anderson, J. Bonaguro, M. McKinney, A. Nicklin, J. Wiseman, “Ethics & algorithms toolkit”, (2019), <<https://ethicstoolkit.ai/>> or Digital Dubai Office, “AI System Ethics Self-Assessment Tool,” (2020) <<https://www.digitaldubai.ae/self-assessment>>.

¹²⁷ “AI Risk Management Framework.” 2021. NIST. <<https://www.nist.gov/itl/ai-risk-management-framework>>.

¹²⁸ Bruschi, Danilo, and Nicla Diomedè, “A Framework for Assessing AI Ethics with Applications to Cybersecurity.” (2023) AI and Ethics 3 (1): 65–72. <<https://doi.org/10.1007/s43681-022-00162-8>>.

¹²⁹ Michelle Seng Ah Lee and Jatinder Singh, “Risk Identification Questionnaire for Detecting Unintended Bias in the Machine Learning Development Lifecycle.” (2021) In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society*, 704–14. AIES '21. New York, NY, USA: Association for Computing Machinery. <<https://doi.org/10.1145/3461702.3462572>>.

3.1. Legal considerations

According to the GDPR and the current wording of the proposal of AIA, both legal acts would be applicable to the BioID AI system.

As the company has been established in the EU, it is within the territorial scope of the GDPR. Additionally, FaceReco also processes personal data as the model is trained on publicly available pictures and databases containing identification data of individuals. The firm in our case would for purposes of training AI systems qualify as the controller of personal data. Furthermore, the essential purpose of the BioID is to allow for the identification of individuals based on the FRT. This means that the personal data in question fulfill the definition of biometric data as provided in GDPR, where “biometric data means personal data resulting from specific technical processing relating to the physical, physiological or behavioral characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopic data.”¹³⁰ Although the bare processing of photographs does not qualify as the processing of biometric data,¹³¹ in our case, FRT shall be characterized as “specific technical processing.”¹³² The mere purpose of the FRT in question is the identification of individuals.¹³³ Identification of an individual by the biometric system has been described by the Article 29 Data Protection Working Party as the “process of comparing biometric data of an individual (acquired at the time of the identification) on several biometric templates stored in a database (i.e. one-to-many matching processes).”¹³⁴ This in practice means that FaceReco has to consider provisions applicable to the processing of specific categories of personal data as biometric data represent sensitive personal data.¹³⁵ Indeed, as highlighted by the European Data Protection Board, in the case of FRT deployed in different spaces, controllers have to satisfy the requirements for specific exceptions for the processing of biometric data.¹³⁶

Taking into account circumstances that trigger the obligation to conduct DPIA, at least two specific criteria are fulfilled. In our view, FRT includes processing on a large scale of special categories of data referred to in Article 9(1)¹³⁷ and specific use of FRT in public spaces may qualify as a systematic monitoring of a publicly accessible area on a large scale.¹³⁸ Considering

the general test of high-risk processing operations,¹³⁹ the case in question encompasses systematic monitoring, processing of sensitive personal data, processing on a large scale, potential matching and combining datasets, and innovative use or applying new technological or organizational solutions where FRT is explicitly mentioned by the WP29. Therefore, FaceReco would definitely be under the obligation to conduct DPIA.

As discussed in the previous part, AIA bans the use of real-time FRT for law enforcement purposes with certain exceptions. Therefore, if BioID would be used by law enforcement, the obligation in question has to be observed.¹⁴⁰ A more pressing issue, in this case, is that the BioID would fall under the high-risk AI systems as it uses machine learning techniques. At the same time, BioID as FRT falls under point 1 of Annex III of the AIA as it shall be classified as “AI systems intended to be used for the ‘real-time’ and ‘post’ remote biometric identification of natural persons.” As the AI system in question represents a high-risk AI system, the obligation to conduct a CA is triggered. Additionally, the CA shall be conducted with the aid of a third party.

3.2. Ethics-based assessment of facial recognition technology

By pursuing a more methodical approach we advocate the ethics-based assessment of FRT not only to encompass the idea of translating the principles of trustworthy AI into a set of specific questions and recommendations but also to understand it as a process of sensitizing providers of FRT to ethical issues considering their AI systems.

We also advocate the idea of a risk-based approach to FRT systems where the possible harms are evaluated by dimensions of exposure, likelihood, and severity and where their prohibition is reasonable when the technology by its design leads to immense risks in regard to these three dimensions that cannot be counter-measured. This approach reflects on a more general debate about the need for red lines in AI¹⁴¹ as opposed to the view that FRT, especially remote FRT, should be prohibited because it poses “unacceptable risks to fundamental rights”¹⁴² without considering the way how to mitigate these risks to avoid or at least minimize possible negative impacts on direct and indirect stakeholders. Some of the most

¹³⁰ GDPR, Article 4 (14).

¹³¹ GDPR, Recital 51.

¹³² See further e.g. S. Solarova, J. Podrouzek, M. Mesarcik, M. et al. “Reconsidering the regulation of facial recognition in public spaces,” (2022) AI Ethics, <<https://doi.org/10.1007/s43681-022-00194-0>>.

¹³³ We consider facial recognition as a means of identification. See Article 29 Data Protection Working Party, “Opinion 3/ 2012 on Developments in Biometric Technologies” WP 193, 27 April 2012, pp. 21-22.

¹³⁴ Article 29 Data Protection Working Party, “Opinion 3/ 2012 on Developments in Biometric Technologies” WP 193, 27 April 2012.

¹³⁵ GDPR, Article 9 (1).

¹³⁶ European Data Protection Board, “Guidelines 3/2019 on the processing of personal data through video devices” (2020). Version 2.0, 29 January 2020).

¹³⁷ GDPR, Article 35 (3) (b).

¹³⁸ GDPR, Article 35 (3) (c).

¹³⁹ GDPR, Article 35 (1).

¹⁴⁰ From the point of the EU data protection law, processing of personal data by FRT used for law enforcement purposes falls under the Law Enforcement Directive. However, the obligation to conduct DPIA also applies in the context of law enforcement purposes pursued by competent authorities as per Article 27 of the Law Enforcement Directive. European Data Protection Board, “Guidelines 05/2022 on the use of facial recognition technology in the area of law enforcement,” (2022). Version 2.0. Adopted on 26 April 2023.

¹⁴¹ Catelijne Muller, “The Impact of Artificial Intelligence on Human Rights, Democracy and the Rule of Law,” (2022) Strasbourg: AD HOC COMMITTEE ON ARTIFICIAL INTELLIGENCE (CAHAI). <<https://rm.coe.int/cahai-2020-06-fin-c-muller-the-impact-of-ai-on-human-rights-democracy-/16809ed6da>>.

¹⁴² See for example a civil society open letter to the CZ Presidency and Ministers working on the EU AI Act. <<https://edri.org/wp-content/uploads/2022/10/CZ-Minister-Digitalisation-letter-AI-act.pdf>> accessed 30.5.2023.

imminent foreseeable impacts of facial recognition technology are already reflected by concerns that are present in the current discussion about the strict regulation or even prohibition of FRT in public spaces.¹⁴³

However, the amount of risk considering facial recognition can also depend on specific use cases or the type of space in which these systems are deployed that need to be taken into account when assessing FRT systems. Such discrepancies are most apparent when analyzing the impacts and effective countermeasures of FRT in open-public and semi-open spaces as airports, stadiums or open streets considering the requirements on transparency, privacy, and human agency.¹⁴⁴ Therefore the ethics-based assessment of facial recognition technology should be sensitive to these circumstances that lead to considerably different levels of risks for stakeholders even the risks of not using the AI system in the specific use case.

The first step in the ethics-based assessment of every FRT system should be the identification of groups and organizations that will be affected by the systems directly or indirectly. In the case of facial recognition systems, we can consider direct stakeholders to be among others the AI providers, system users, persons intentionally captured by FRT, or persons who are part of the biometric databases for the sake of their approval to access the space under surveillance. However, stakeholder analysis is capable of disclosing also many indirect stakeholder groups. Among many others, the third-party operators, uninformed captured persons, video source owners, or people whose faces were used to train AI models could be also considerably affected, especially by the widespread or long-term use of facial recognition systems.

The evaluation of direct and indirect stakeholders and their classification into different categories regarding the estimation of the degree of impacts and the degree of their awareness about these impacts provides additional value in terms of better identification of possible risks for rights, freedoms, and interests of stakeholders but also helps to better understand their vulnerability towards these risks. From this point of view, the persons that were intentionally captioned and who are fully aware of that fact will be probably less vulnerable than persons that are ignorant of such activities even if the degree of impact from FRT would be similar to them. Although such estimations should be further validated with stakeholders, the transparent and open communication strategy from FRT providers based on segmentation from the stakeholder matrix can lead to better and more appropriate engagement with stakeholders. Especially regarding the indirect but still considerably vulnerable stakeholders, like people whose data were used for training the FRT system. However, such estimations should not be exercised solely by providers of FRT systems and need to be validated in participation with most affected groups.

In practice, stakeholder analysis as part of the ethics-based assessment of FRT offers a deeper understanding of technology implementation than seeking views of data subjects ac-

cording to GDPR during DPIA. Additionally, as discussed earlier, this part of DPIA may be skipped if the controller argues that such a process would be inappropriate. Taking into account potential stakeholders may also be part of the post-market monitoring according to the AIA, where the provider of high-risk AI systems shall “actively and systematically collect, document and analyze relevant data provided by users or collected through other sources on the performance of high-risk AI systems throughout their lifetime.”¹⁴⁵ However, mere compliance with the aforementioned regulations may not reveal the whole picture of affected subjects and potential impacts on them.

Also, the map of data flows and assessment of data sources used by FRT systems can help with the identification of practices that, although they are legal, still raise questions of their tenability from a moral perspective. Some data sources may be questioned regarding their reliability and fair use especially when FRT systems were trained on data crawled from the internet. The existence of similar practices is apparent in cases where data is acquired from countries with lower levels of personal data protection, or concerning the data collected from marginalized groups such as crime offenders, children, or even deceased people.¹⁴⁶ Currently, data flow mapping as described above is not an explicit legal obligation. However, drafting a data map may be a part of “the systematic description of processing” pursuant to Article 35 (7) a) GDPR or data governance according to Article 10 of the AIA. Here, more complex mapping of data flows may reveal unexpected subjects involved in data processing and ethically questionable practices that shall be taken into consideration.

Considering the ethical or societal issues that may emerge during the ethics-based assessment, we are offering a few more examples based on ALTAI questions. The requirement of human agency and oversight regarding trustworthy AI systems opens up the discussion about possible misuse or overuse of facial recognition that goes beyond actual regulation. Besides the risk of unauthorised access by unknown or unauthorised persons, AI providers should face the risk that even authorized users will tend to over-rely on facial recognition technology. The impacts of lower accuracy of facial recognition systems could be severe, but combined with average high accuracy in standard conditions can lead to the false impressions of some users that the “machine is always right.”

The long-term effects of blind faith in automated identification that result in automation bias¹⁴⁷ are still to be evaluated but there are concerns that identification without human intervention or oversight may lead to the weakening of human responsibility or even the erosion of human auton-

¹⁴⁵ AIA, Article 61 (2).

¹⁴⁶ Lydia. n.d. Morrish, “A Face Recognition Site Crawled the Web for Dead People’s Photos.” *Wired*, <<https://www.wired.com/story/a-face-recognition-site-crawled-the-web-for-dead-peoples-photos/>> accessed 30.5.2023.

¹⁴⁷ Mark Sujan, Dominic Furniss, Kath Grundy, Howard Grundy, David Nelson, Matthew Elliot, Sean White, Ibrahim Habli, and Nick Reynolds, “Human Factors Challenges for the Safe Use of Artificial Intelligence in Patient Care,” (2019) *BMJ Health Care Inform.* 2019 Nov;26(1):e100081, doi: <[10.1136/bmjhci-2019-100081](https://doi.org/10.1136/bmjhci-2019-100081)>. PMID: 31780459; PMCID: PMC7252977>.

¹⁴³ See e.g. ALLAI, “Council of Europe Calls for a Ban on Certain Facial Recognition Applications–ALLAI,” <<https://allai.nl/council-of-europe-calls-for-a-ban-on-certain-facial-recognition-applications/>> (2021) accessed 25.10.2022.

¹⁴⁴ *Ibid.*

omy.¹⁴⁸ The inappropriate evaluation of decisions made by a system or relying too much on the decisions made by an AI system in long-term use may weaken the human autonomy of stakeholders and their ability to form their own decisions. The aspect of over-reliance or human autonomy is currently fairly unregulated by law. Although Article 14 of the AIA will require subjects in the scope of the regulation to implement human oversight for high-risk AI systems until the act comes into force these aspects are currently not covered in its complexity.

It has to be mentioned that there are multiple levels of abstraction when it comes to thinking about what constitutes a relevant stakeholder. Some approaches include nonhuman animals¹⁴⁹ or even non-living entities such as the environment, and relatively general stakeholder groups such as “the society.” This also applies to the ALTAI, which discusses the society and environment as relevant stakeholders within the AI life cycle. As the whole society could be considered a stakeholder affected by the use of FRT,¹⁵⁰ we should think about FRT systems not only as technological artifacts by themselves but always conceptualize them in a broader context with specific social norms and values affected by their use. There are numerous examples of when FRT is misused by organizations or non-democratic regimes for public surveillance.¹⁵¹ We should be sensitive to the public fears regarding the risks of public surveillance to avoid the chilling effect of FRT. Some of these risks concern the omnipresence of FRT in public spaces where people are forced to biometric identification even without their awareness and consent. Therefore, the chilling effect and broader societal impacts remain a severe risk from legal and ethical points of view.

However, the developers and deployers as direct stakeholders could be negatively affected by FRT, too. They risk their trustworthiness and credibility not only as a part of the organization they form but also as human beings that are accountable for the development and deployment of FRT systems. Similar to other AI professionals, they are struggling when they need to cope with the moral dilemmas that question their professional loyalty when having concerns about the considerable harm to society that may be caused by the system they are working on.¹⁵²

Overall, the problem of accountability should be understood as one of the most important ethical issues of FRT and also of AI systems in general. The problem of many hands¹⁵³ that leads to responsibility gaps opens the question of who

should be deemed responsible for decisions of AI systems. We defend the view that the responsibility roles should be prescribed before the system is deployed and ready for use. The application of ex-ante or forward-looking accountability models during ethics-based assessment has the ambition to prescribe such responsibility as a duty of reasonable care to avoid foreseeable harm to others.¹⁵⁴ Therefore the risk matrix for FRT systems should contain the list of countermeasures to ethical risks together with the roles that will be accountable for implementing these countermeasures and clearly state who can be deemed responsible when the countermeasures were not effectively put in place.

Additionally, we believe that such an approach concerning the analysis of various types of risks is in line with GDPR's implementation of principles of data protection by design and by default according to Article 25. It shall be noted that privacy shall not be perceived as a pure regulatory risk or issue, but is often discussed also from the point of view of ethics.¹⁵⁵ EDPB explicitly mentions ethical considerations as part of the fairness principle and subsequent compliance with Article 25 GDPR.¹⁵⁶ Ethical considerations under GDPR may also stem from the interpretation of the principle of accountability and provide space for corporate social responsibility.¹⁵⁷ AIA will require ex-ante assessment and ethics-based assessment may significantly aid developers, deployers, and users of AI systems to evaluate different types of risks associated with the use of the system. Following only legal requirements or conducting an assessment of the impact on rights, and freedoms of interests from a legal point of view might miss important ethical and societal risks. This is also evident from the evolving framework of human rights, ethical and social impact assessment.¹⁵⁸ Ethical implications of new technologies represent a broader societal context that calls for further considerations than privacy and data protection assessments offered.¹⁵⁹ Not only individuals but rather groups and society shall also be taken into account concerning the impact of new technologies.¹⁶⁰

Based on the findings above, the answer to our research question is based on two significant arguments. Ethics-based

¹⁴⁸ A. Marciano, “Reframing biometric surveillance: from a means of inspection to a form of control,” (2019) *Ethics Inf. Technol.* 21(2), 127–136. <<https://doi.org/10.1007/s10676-018-9493-1>>.

¹⁴⁹ Peter Singer and Yip Fai Tse, “AI Ethics: The Case for Including Animals.” (2022) *AI and Ethics*, July. <<https://doi.org/10.1007/s43681-022-00187-z>>.

¹⁵⁰ S. Zuboff, “The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power,” (2016), Profile Books, London.

¹⁵¹ R. Creemers, “China’s social credit system: an evolving practice of control,” (2018), SSRN, <<https://doi.org/10.2139/ssrn.3175792>>.

¹⁵² C. Miller and R. Coldicutt People, “Power and Technology: The Tech Workers’ View,” (2019), London: Doteveryone.

¹⁵³ D.F. Thompson, “Moral responsibility and public officials,” (1980) *American Political Science Review*, 74, 905–916.l.

¹⁵⁴ N. Doorn, “Responsibility Ascriptions in Technology Development and Engineering: Three Perspectives,” *Sci Eng Ethics* 18, 69–90 (2012 <<https://doi.org/10.1007/s11948-009-9189-3>>.

¹⁵⁵ David Wright, Michael Friedewald, “Integrating privacy and ethical impact assessments” *Science and Public Policy*, Volume 40, Issue 6, December 2013, pp. 755–766, <<https://doi.org/10.1093/scipol/sct083>>

¹⁵⁶ European Data Protection Board, “Guidelines 4/2019 on Article 25 Data Protection by Design and by Default,” (2019), Version 2.0. Adopted on 20 October 2020.

¹⁵⁷ H. Hijmans & C. Raab, “Ethical Dimensions of the GDPR, AI Regulation, and Beyond”, (2022) *Direito Público*, 18(100). <<https://doi.org/10.1111/rdp.v18i100.6197>>.

¹⁵⁸ Alessandro, Mantalero, “Beyond DataHuman Rights, Ethical and Social ImpactAssessment in AI,” (2022) T.M.C.ASSER PRESS, The Hague, The Netherlands;

¹⁵⁹ Charles D. Raab, “Information privacy, impact assessment, and the place of ethics,” *Computer Law & Security Review* Volume 37, (2020), 105404, <<https://doi.org/10.1016/j.clsr.2020.105404>>.

¹⁶⁰ David, Wright - Charles D. Raab, “Constructing a surveillance impact assessment” *Computer Law & Security Review* Volume 28, Issue 6, December 2012, pp. 613–626.

assessments may improve compliance with EU digital laws by accommodating ethical considerations within a narrower set of regulatory obligations. Although some legal requirements are voluntary or implicit, the approach described in this part is suitable for compliance and beyond. Additionally, analysis of broader societal and ethical risks also improves the trustworthiness of new technologies as emerging risks may be mitigated.

4. Conclusions

The article is presented as a direct response to defend the role of ethics and its operationalization in ICT governance as they are gaining more academic and practical attention. Our aim was to reflect on the positive side of ethicification and show valuable insights that ethics may bring to the regulation of the digital space. We perceive the positive side of ethicification mainly in two manners.

Firstly, ethics-based assessments fit into the existing and proposed regulatory requirements containing obligations for ex-ante assessments. We are convinced that DPIA pursuant to Article 35 GDPR and CA according to the proposal of AIA are well suited to accommodate ethics-based assessments in the process. However, the limited scope (processing of personal data with high-risk and high-risk AI systems) shall be emphasized as many personal data practices and AI systems may escape the scrutiny provided by these obligations.

Secondly, ethics-based assessments bring additional value to the already existing regulatory obligations. Specifically, precise stakeholder analysis helps to identify a wider range of relevant subjects influenced by the processing of personal data or AI systems. This results in a more thorough identification of risks that might have been missed within pure compliance. Furthermore, drafting a data flow map may reveal ethical issues related to used datasets and data sources beyond regulatory requirements. These steps are essential for the identification of broader societal and ethical risks and for providing countermeasures to mitigate them. On the examples of hu-

man autonomy, chilling effect, and misuse by third parties we have elaborated specifics of such risks and their need to be mitigated.

We also perceive conducting ethics-based assessments as an opportunity to mitigate the gap until EU AI regulation comes into force. Risks associated with AI systems will not magically emerge after the adoption of the legislation but represent a current challenge for various stakeholders. Taking into account a broader range of risks also aids in improving the trustworthiness of AI systems operated in regulatory gray zones. Additionally, the integration of consideration of legal, ethical, and societal risks serves as an example par excellence in how to involve different modalities of regulation into practice.

We acknowledge the limitations of ethics-based assessments in general including ethics shopping or ethics bluewashing. The process requires ethical professionals with a deeper understanding of moral concepts with respect to specific uses of technology who can help AI providers in terms of better ethical sensitivity during the development and deployment process. At the same time, we are admitting the limitations of the approach used in the use case scenario that is directly influenced by specific context and circumstances. However, our main points remain valid. We believe that our contribution to the discourse on the ethicification of law will invite other professionals and academics to express their position on this important topic.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.